

扩展题

2024.2.22

Name: Chen Zhuoyan Student ID: 523030910037

第二章扩展题

1.

$$1 \times 8^2 + 7 \times 8 + 6 \times 8^0 + 5 \times 8^{-1} + 1 \times 8^{-2} = 126.640625$$

2.

Remainder

$$173/2 = 86 \quad 1$$

$$86/2 = 43 \quad 0$$

$$43/2 = 21 \quad 1$$

$$21/2 = 10 \quad 1$$

$$10/2 = 5 \quad 0$$

$$5/2 = 2 \quad 1$$

$$1/2 = 0 \quad 1$$

$$\therefore 173 \rightarrow 1101101$$

Carry

$$0.39 \times 2 = 0.78 \quad 0$$

$$0.78 \times 2 = 1.56 \quad 1$$

$$\therefore 0.39 \rightarrow 0.01$$

$$\therefore 173.39 \rightarrow 1101101.01$$

$$3.1011110.1011001 \rightarrow 1|011|110.101|100|1$$

$$\therefore (1011110.1011001)_2 = (136.544)_8$$

$$1011110.1011001 \rightarrow 0101|1110.1011|0010$$

$$\therefore (1011110.1011001)_2 = (5E.B2)_{16}$$

4.

Remainder

$$87/8 = 10 \quad 7$$

$$10/8 = 1 \quad 2$$

$$1/8 = 0 \quad 1$$

$$\therefore (87)_{10} = (127)_8$$

Remainder

$$87/16 = 5 \quad 7$$

$$5/16 = 0 \quad 5$$

$$\therefore (87)_{10} = (57)_{16}$$

$$5.(1) \text{ 原码: } 01011$$

$$1\text{'s complement: } 01011$$

$$2\text{'s complement: } 01011$$

$$(2) \text{ 原码: } 11101$$

$$1\text{'s complement: } 10010$$

$$2\text{'s complement: } 10011$$

6.

	Remainder
$121/2 = 60$	1
$60/2 = 30$	0
$30/2 = 15$	0
$15/2 = 7$	1
$7/2 = 3$	1
$3/2 = 1$	1
$1/2 = 0$	1

$\therefore (+121)_{10} = (01111001)_2$
 $\therefore 1$'s complement of 01111001 is 10000110
 $\therefore 2$'s complement of 01111001 is 10000111

7.(1)
 $(3)_{10} = (0000011)_2$
 $(15)_{10} = (0001111)_2$
 $\therefore 2$'s complement of 3 and 15 are 00000011 and 00001111

$\therefore$

$$\begin{array}{r} 00000011 \\ + 00001111 \\ \hline 00010010 \end{array}$$
 $\therefore (3 + 15)_{10} = (00010010)_2 = (18)_{10}$

(2)
 $(12)_{10} = (0001100)_2$
 $(7)_{10} = (0000111)_2$
 $\therefore 2$'s complement of 12 and -7 are 00001100 and 11111001

$\therefore$

$$\begin{array}{r} 00001100 \\ + 11111001 \\ \hline 10000101 \end{array}$$
 $\therefore (12 - 7)_{10} = (00000101)_2 = (5)_{10}$

(3)
 $(12)_{10} = (0001100)_2$
 $(5)_{10} = (0000101)_2$
 $\therefore 2$'s complement of -12 and -5 are 11110100 and 11111011

$\therefore$

$$\begin{array}{r} 11110100 \\ + 11111011 \\ \hline 11110111 \end{array}$$
 $\therefore (-12 - 5)_{10} = (11101111)_2 = (-17)_{10}$

8.
 $(9)_{10} = (1001)_{BCD}$
 $(3)_{10} = (0011)_{BCD}$

$\therefore$

$$\begin{array}{r} 1001 \\ + 0011 \\ \hline 1100 \\ + 0110 \\ \hline 1\ 0010 \end{array}$$

$$\therefore 1001 + 0011 = 1|0010 = (12)_{10}$$

9. $\therefore$ There are 6 1s in 110001010101 and there are 3 1s in 1110000
 $\therefore$ 1110000 is wrong.

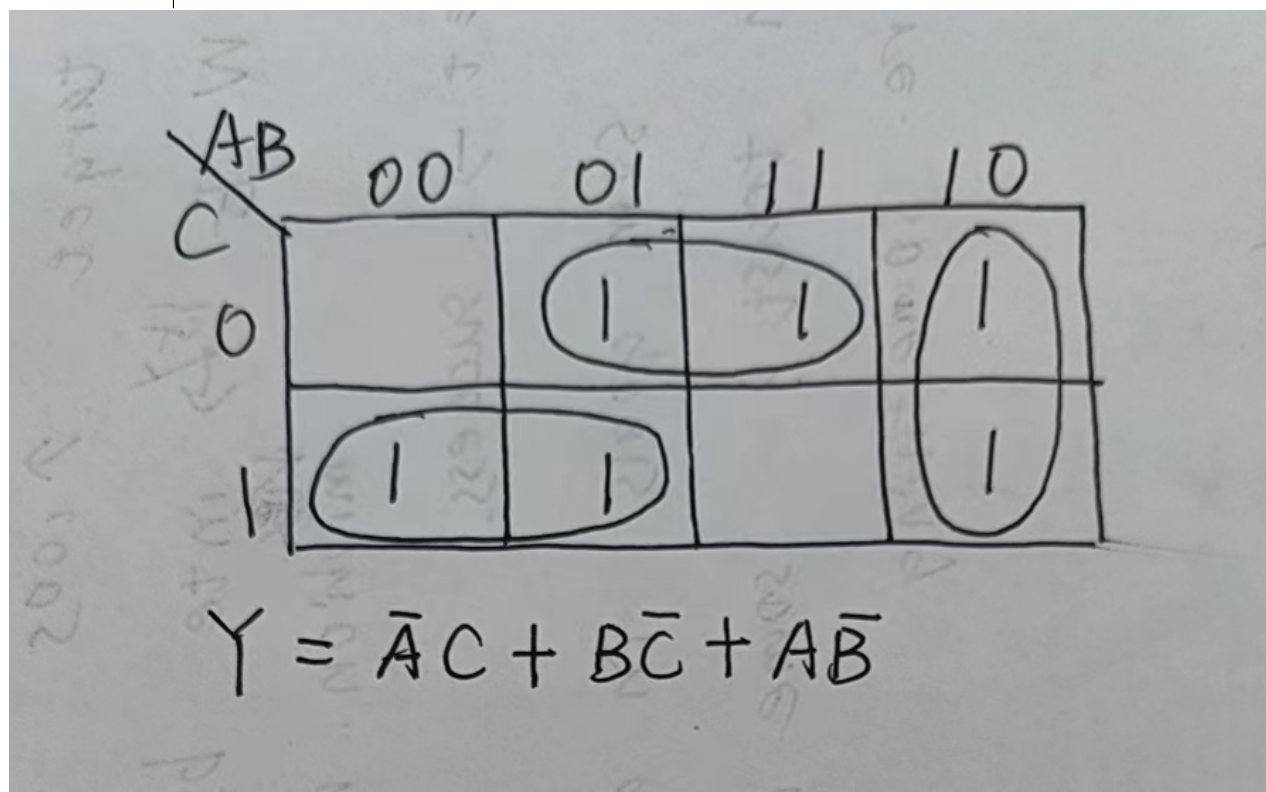
10. (1)B.D (2)D. (3)D.

第四章扩展题:

$$1. Y = AB + \overline{A}B + B\overline{C} + \overline{B}C$$

Truth table:

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0



$$\begin{aligned}
 2. (1) Y &= AB(\overline{A}CD + \overline{(AD + \overline{B} \cdot \overline{C})})(\overline{A} + B) \\
 &= AB(\overline{A} + B)(\overline{A}CD + \overline{(AD + \overline{B} \cdot \overline{C})}) \\
 &= (A\overline{A} \cdot \overline{B} + AB\overline{B})(\overline{A}CD + \overline{(AD + \overline{B} \cdot \overline{C})}) \\
 &= (0 + 0)(\overline{A}CD + \overline{(AD + \overline{B} \cdot \overline{C})})
 \end{aligned}$$

$$= 0$$

$$\begin{aligned}(2) Y &= AC(\overline{C}D + \overline{A}B) + BC(\overline{B} + AD) + CE \\ &= (AC\overline{C}D + A\overline{A}BC) + BC((\overline{B} + AD)(\overline{C} + \overline{E})) \\ &= (0 + 0) + (B\overline{B} + ABD)(\overline{C}\overline{C} + C\overline{E}) \\ &= ABCDE\overline{E}\end{aligned}$$

$$\begin{aligned}(3) Y &= A + \overline{B} + \overline{C}(A + \overline{B} + C)(A + B + C) \\ &= A + \overline{B}C((A + C)(A + C) + (A + C)(B + \overline{B}) + B\overline{B}) \\ &= A + \overline{B}C(A + C) \\ &= A + A\overline{B}C + \overline{B}C \\ &= A(1 + \overline{B}C) + \overline{B}C \\ &= A + \overline{B}C\end{aligned}$$

$$\begin{aligned}(4) Y &= B\overline{C} + AB\overline{C}E + \overline{B}(\overline{A} \cdot \overline{D}) + AD + BC(\overline{A}\overline{D} + \overline{A}D) \\ &= B\overline{c}(1 + AE) + \overline{B}(A + D)(\overline{A} + \overline{D}) + B(\overline{A}\overline{D} + \overline{A}D) \\ &= B\overline{C} + \overline{B}(\overline{A}\overline{D} + \overline{A}D) + B(\overline{A}\overline{D} + \overline{A}D) \\ &= B\overline{C} + (\overline{A}\overline{D} + \overline{A}D)(B + \overline{B}) \\ &= B\overline{C} + \overline{A}\overline{D} + \overline{A}D\end{aligned}$$

$$\begin{aligned}3.(a) Y &= \overline{\overline{A\overline{B}C} \cdot B\overline{C}} \\ &= \overline{A\overline{B}C + B\overline{C}}\end{aligned}$$

$$\begin{aligned}(b) Y_1 &= \overline{AB + C(A \oplus B)} \\ &= \overline{AB + C(\overline{A}B + A\overline{B})} \\ &= \overline{AB + A\overline{B}C + \overline{A}BC} \\ &= \overline{A(B + \overline{B}C) + \overline{A}BC} \\ &= \overline{A(B + C) + \overline{A}BC} \\ &= \overline{AB + (A + \overline{A}B)C} \\ &= \overline{AB + (A + B)C} \\ &= \overline{AB + AC + BC} \\ Y_2 &= (A \oplus B) \oplus C \\ &= (\overline{A\overline{B} + \overline{A}B})\overline{C} + (\overline{A\overline{B} + \overline{A}B})C \\ &= \overline{A\overline{B} \cdot \overline{C}} + \overline{A\overline{B}C} + (\overline{A} + B)(A + \overline{B})C \\ &= \overline{A\overline{B} \cdot \overline{C}} + \overline{A\overline{B}C} + ABC + \overline{A} \cdot \overline{B}C\end{aligned}$$

$$4. Y = \overline{A}BC + AC + \overline{B}C$$

Truth table:

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

$$\therefore Y = \overline{A} \cdot \overline{B}C + \overline{A}BC + A\overline{B}C + ABC$$

$$Y = (A + B + C)(A + \overline{B} + C)(\overline{A} + B + C)(\overline{A} + \overline{B} + C)$$

$$5. Y = \Sigma m(1, 4, 5, 6, 7, 8, 10, 14)$$

AB \ CD	00	01	11	10
00		1	ϕ	ϕ
01	1	1	1	1
11				1
10	1			1

$$\therefore Y = \overline{A}B + \overline{A} \cdot \overline{C}D + A\overline{B} \cdot \overline{D} + AC\overline{D}$$

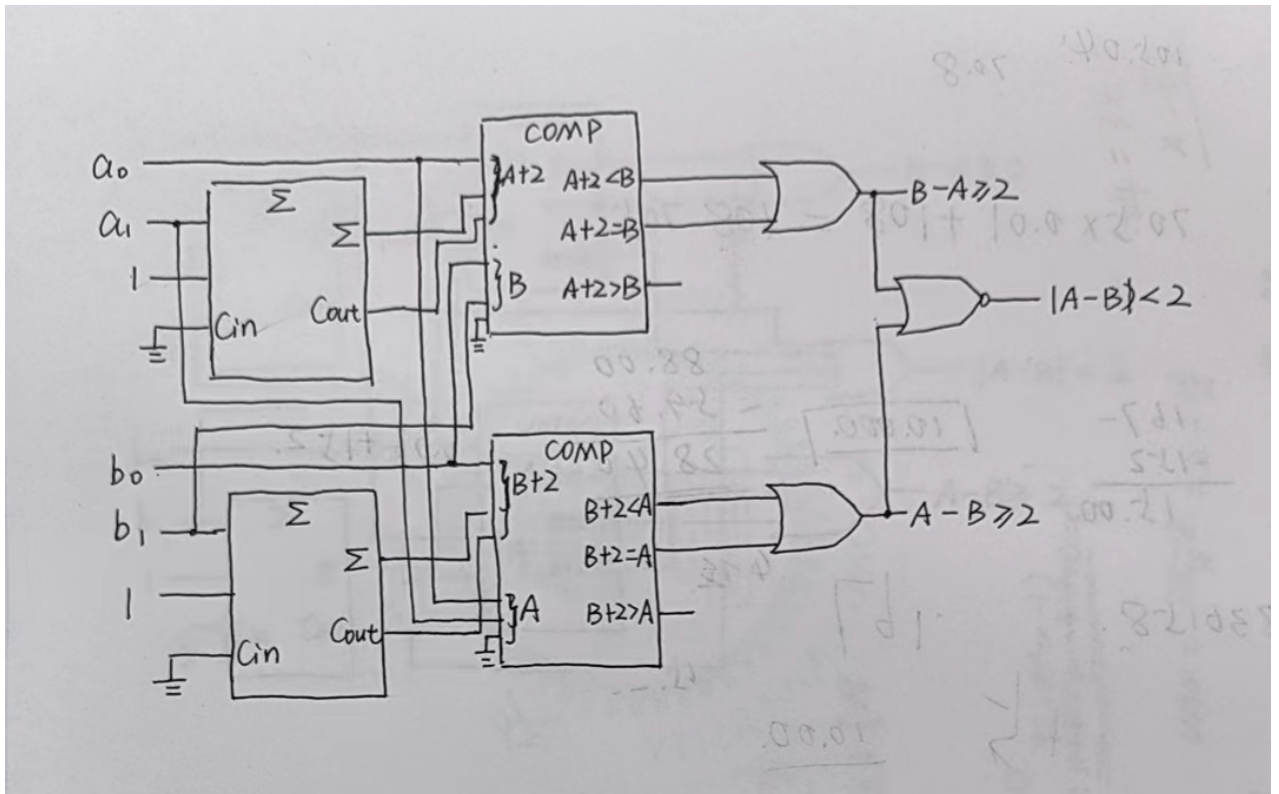
$$\therefore Y = \Sigma m(1, 4, 5, 6, 7, 8, 10, 14) = \Pi m(0, 2, 3, 9, 11, 12, 13, 15)$$

AB \ CD	00	01	11	10
00	0		0	0
01				
11	0	0	0	
10		0	0	

$$\therefore Y = (\overline{A} + \overline{D})(\overline{A} + \overline{B} + C)(A + B + D)(A + B + \overline{C})$$

第六章扩展题:

1.



$$\begin{aligned}
 2. Z = Y_1 + Y_2 &= (0 \cdot \bar{N} \cdot \bar{M} + 0 \cdot \bar{N} M + Q N \bar{M} + Q M N) \cdot \bar{P} + (Q \cdot \bar{N} \cdot \bar{M} + Q \cdot \bar{N} M + 0 \cdot N \bar{M} + 0 \cdot M N) \cdot P \\
 &= \bar{P}(Q N (\bar{M} + M)) + P(Q \bar{N} (M + \bar{M})) \\
 &= \bar{P} Q N + P Q \bar{N}
 \end{aligned}$$

3. Truth table:

A	B	C	Y
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

$$\therefore F = \bar{A} \cdot \bar{B} \cdot \bar{C} + \bar{A} B C + A B \bar{C}$$